Actions Taken

- Enabled Multi-Factor Authentication for all IAM users.
- Configured and verified a least-privilege IAM role (SecurityAdmin) following the principle of least privilege.
- Enabled and configured AWS CloudTrail, with integration to CloudWatch Logs for monitoring and alerting.
- Applied security controls to an S3 bucket, including public access blocking, default server-side encryption, server access logging.

AWS Security Hardening

AWS operates under the Shared Responsibility Model: AWS secures the cloud infrastructure, while customers are responsible for security in the cloud.

- It reduces the attack surface and prevents the majority of breaches caused by misconfiguration.
- It enforces zero-trust principles and compliance requirements.
- It protects sensitive data in S3 and limits the impact of compromised credentials.

Summary of Improvements Made

The implemented controls deliver measurable security gains:

- MFA enforcement helps block unauthorized access even if passwords are compromised.
- Least-privilege limits permissions to only what is required, reducing risk of privilege escalation.
- CloudTrail/CloudWatch provides complete audit logs and monitoring, enabling detection of anomalous IAM activity and automated alerts.
- S3 security enhancements eliminate public exposure, enforce encryption by default, log all requests, and apply explicit bucket policies, preventing data leaks and unauthorized access.

Enable MFA for Admin Users

Type	Identifier	Certifications	Created on
Virtual	arn:aws:iam::386707340940:mfa/MyDevice	Not Applicable	Sun Apr 19 2026

Least-Privilege IAM Role

Role name	Trusted entities	Last activity
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Se	11 minutes ago
AWSServiceRoleForSupport	AWS Service: support (Service-Linke	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	-
SecurityAdmin	AWS Service: ec2	-

S3 bucket with security policy

Successfully edited bucket policy.

Copy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Principal": {
        "AWS": "*"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::hackme-im-vulnerable/*",
      "Condition": {
        "Bool": {
          "aws:SecureTransport": "false"
        }
      }
    }
  ]
}
```

AWS CloudTrail and CloudWatch

General details

Edit

Trail logging Logging	**Trail log location** hackme-im-vulnerable/AWSLogs/386707340940	**Log file validation** Enabled	**SNS notification delivery** Disabled
Trail name Security-Trail	**Last log file delivered** -	**Last file validation delivered** -	**Last SNS notification** -
Multi-region trail Yes	**Log file SSE-KMS encryption** Not enabled		
Apply trail to my organization Not enabled			

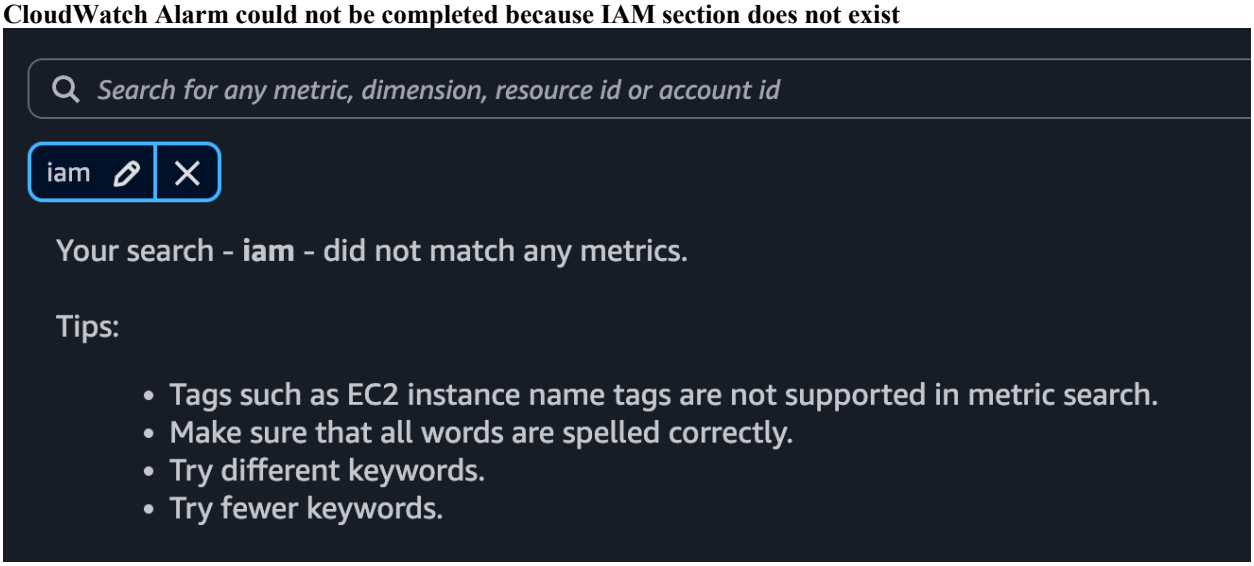
CloudWatch Logs

Edit

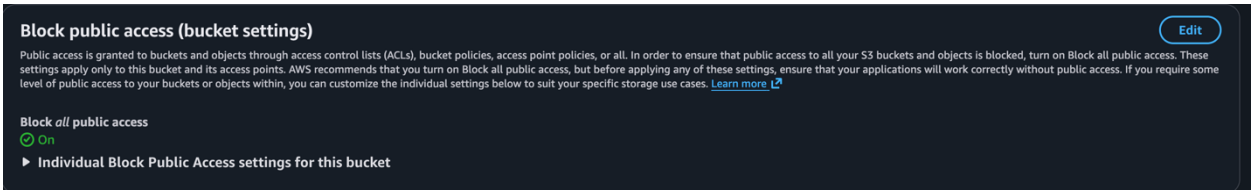
Log group
/aws/cloudtrail/SecurityLogs

IAM Role
arn:aws:iam::386707340940:role/service-role/CloudTrailRoleforCloudWatchLog

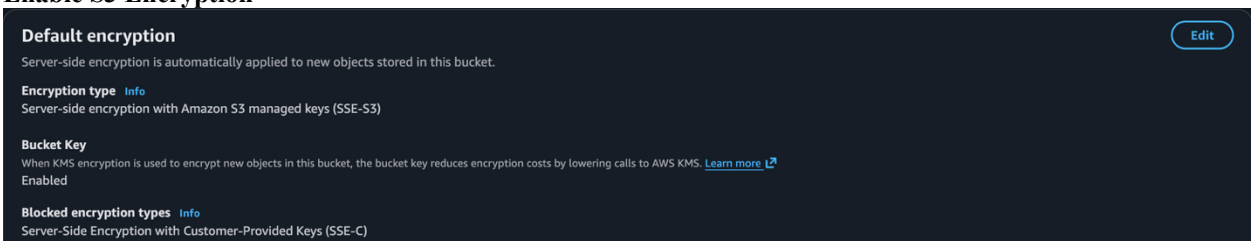
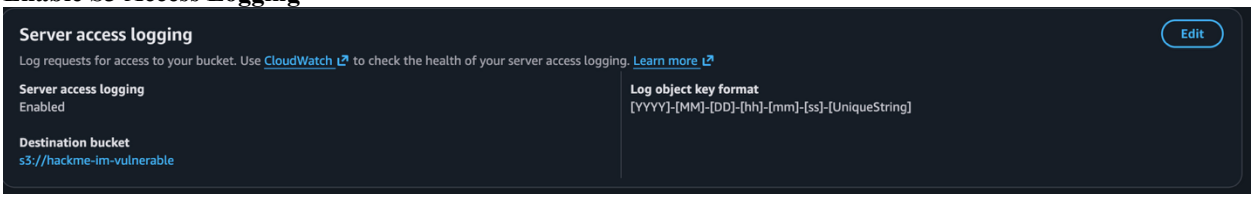
CloudWatch Alarm could not be completed because IAM section does not exist



Restrict Public Access to S3 Buckets



Enable S3 Encryption



Enable S3 Access Logging